

Bonus task 3 5.10.2026

Beam structure

Deadline: Before the lectures on the 12th of October

Possibility to earn five bonus points in total:

- **2 points** from the submitted MATLAB/Python assignment and report.
- **3 points** from short pen-and-paper quizzes during the lecture.

The purpose of this bonus task is to solve the vertical displacement u_Y and rotation θ_Z of the beam structure using the finite element method. Determine the maximum vertical displacement of the structure and its location along the beam. Furthermore, calculate the member forces for all beam elements and construct the bending moment diagram based on the finite element solution.

Investigate the effect of mesh refinement by systematically increasing the number of finite elements and compare the maximum vertical displacement obtained with different meshes. Compare the finite element results with the analytical solution and explain the reasons for any differences or similarities.

Use either MATLAB or Python based on your programming language selection on the beginning of the course. Use the provided MATLAB/Python code templates as a starting point for the finite element solution.

Use

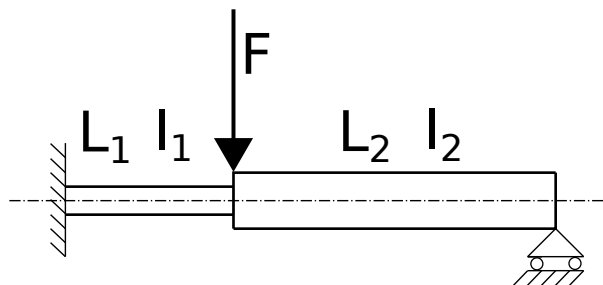
`run_LinStaticBeam_BonusTask3_solution_v10.m` (MATLAB)

or

`run_LinStaticBeam_BonusTask3_solution_v10.py` (Python)

as the starting point for the finite element solution.

The provided solution file uses one Euler-Bernoulli beam element for each beam section, i.e. two finite elements for the whole structure. Modify the code so that a larger number of finite elements can be used. Use the modified code to perform the mesh refinement study and investigate the effect of increasing the number of elements on the finite element solution.



The parameters of the problem are given in the following table.

	Young's modulus, E [MPa]	Second moment of area, I [mm ⁴]	Length, L [mm]
Beam 1	210000	4762	1000
Beam 2	210000	Student-specific I_2	2000
Force, F [N]	100		

The second moment of area of Beam 2 depends on the last digit of your student number. Let n_{student} denote the last digit of your student number, where

$$n_{\text{student}} \in \{0, 1, \dots, 9\}. \quad (1)$$

The second moment of area of Beam 2 is defined as

$$I_2 = 6000 + 500 n_{\text{student}} \quad [\text{mm}^4]. \quad (2)$$

Use the value of I_2 corresponding to your own student number throughout the calculations, finite element analysis, convergence study, and comparison with the analytical solution.

The maximum score of this bonus task is **5 points**. The points are divided into the submitted work and an in-class quiz.

Submitted report and MATLAB/Python code: 2 points

- **Finite element implementation and solution (1 point).**

The Euler-Bernoulli beam element is correctly implemented, the global system is assembled correctly, the boundary conditions are applied properly, and the submitted MATLAB/Python code runs without errors. The main finite element results, including nodal displacements, rotations, and member forces, are reported.

- **Analysis and interpretation of the results (1 point).**

Determine the maximum vertical displacement and its location, construct the bending moment diagram, investigate the effect of mesh refinement, and compare the finite element results with the analytical solution. Briefly explain the observed differences or similarities.

In-class quiz: 3 points

A short individual quiz (approximately 20 minutes) will be organized during the coming lectures. The quiz is completed using pen and paper and tests your understanding of the concepts used in this bonus task.

The quiz may include questions related to:

- Euler-Bernoulli beam interpolation and shape functions,
- elemental stiffness matrix and internal forces,
- global assembly and boundary conditions,
- interpretation of nodal displacements and rotations,
- member forces and bending moments,
- mesh refinement and convergence,
- comparison between the finite element and analytical solutions.

The quiz is worth a maximum of **3 points**.

Tips for an elemental stiffness matrix (done partly during lecture)

Two node 4 DOF Euler-Bernoulli type beam element has displacement v_1 and rotation θ_1 at node 1 ($x = 0$) and v_2 and rotation θ_2 at node 2 ($x = L$). So it is assumed that the element's local coordinate system $\{x, y\}$ is at an element's node 1 ($x = 0$).

Four polynomials are needed to approximate the displacement field v_h in the vertical direction over the element's length (material coordinate x).

$$v_h = a_0 + a_1x + a_2x^2 + a_3x^3 \quad (3)$$

Rotation θ can be approximated from the displacement field as follows:

$$\theta \approx \frac{dv_h}{dx} = a_1 + 2a_2x + 3a_3x^2 \quad (4)$$

Then giving boundary conditions at elemental level $v_h(x = 0) := v_1$, $\theta(x = 0) := \theta_1$, $v_h(x = L) := v_2$ and $\theta(x = L) := \theta_2$:

$$\begin{cases} v_h(0) := a_0 = v_1 \\ \theta(0) := a_1 = \theta_1 \\ v_h(L) := a_0 + a_1L + a_2L^2 + a_3L^3 = v_2 \\ \theta(L) := a_1 + 2a_2L + 3a_3L^2 = \theta_2 \end{cases} \quad (5)$$

Eqs. (5) in a matrix form:

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & L & L^2 & L^3 \\ 0 & 1 & 2L & 3L^2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}}_{\mathbf{a}} = \underbrace{\begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix}}_{\mathbf{u}} \quad (6)$$

then solving $\mathbf{a} = \mathbf{A}^{-1}\mathbf{u}$ and substituting them into Eq.(3).

$$v_h = (\mathbf{A}^{-1}\mathbf{u})^T \mathbf{p} = \underbrace{\mathbf{p}^T \mathbf{A}^{-1}}_{\mathbf{N}} \mathbf{u} \quad (7)$$

where $\mathbf{N}$ includes shape functions (known as ansatz functions in German literature).

Strain energy and elemental stiffness matrix

Due to the kinematic assumptions in Euler-Bernoulli theory, vertical displacement u at distance y from the beam's neutral axis can be written as:

$$u(x) = -y \frac{\partial v}{\partial x} \quad (8)$$

and strain

$$\varepsilon = \frac{\partial u}{\partial x} = -y \frac{\partial^2 v}{\partial x^2} = -y \underbrace{\frac{\partial \theta}{\partial x}}_{\varepsilon_{xx}^1} \quad (9)$$

Using the assumptions, strain energy can be written as:

$$W_{int} = \frac{1}{2} \int_V \sigma \varepsilon dV = \frac{1}{2} E \int_0^L \varepsilon_{xx}^1{}^2 \int_A y^2 dA dx = \frac{1}{2} EI \int_0^L \varepsilon_{xx}^1{}^2 dx \quad (10)$$

where $I = \int_A y^2 dA$ is moment of inertia and ε_{xx}^1 is curvature. Curvature (kinematic measure $\varepsilon_{xx}^1 = 1/\text{radius of a curved beam}$) tells how much the beam has deformed by bending moment and it is approximately equal to the second derivative of the deflection with respect to material coordinate x :

$$\varepsilon_{xx}^1 \approx \frac{d\theta}{dx} = \frac{d^2 v_h}{dx^2} \quad (11)$$

The vector of elemental internal forces:

$$\mathbf{f}_{int} = \frac{\partial W_{int}}{\partial \mathbf{u}} \quad (12)$$

Elemental local stiffness matrix:

$$\bar{\mathbf{K}} = \frac{\partial \mathbf{W}}{\partial \mathbf{u}} \quad (13)$$

Analyzed structures are not often parallel to the element's longitudinal direction -> Transformation from a local elemental coordinate system to a global coordinate system.

PLEASE NOTE: In this task, the element's local coordinate system is parallel to the beam's global coordinate system, so no transformation is required. You could still perform one, but the element-level global stiffness matrix would then contain zero rows and columns that must be eliminated.

Assembling

Sometimes is helpful to present elements' connectivity and corresponding nodal displacements in a table as follows:

	Node I	Node J	Global nodal displacements	Local nodal displacements
Beam 1	1	2	$u_{X_1}, \theta_{Z_1}, u_{X_2}, \theta_{Z_2}$	$v_1, \theta_1, v_2, \theta_2$
Beam 2	2	3	$u_{X_2}, \theta_{Z_2}, u_{X_3}, \theta_{Z_3}$	$v_3, \theta_3, v_4, \theta_4$

It can be seen from the connectivity table what stiffness components of global elemental stiffness matrices $\mathbf{K}_G^{(1)}$ and $\mathbf{K}_G^{(2)}$ are needed to sum. To make assembling, you need to keep in mind that the global stiffness matrix $\mathbf{K}_G$ for the analysed beam structure should be a size of 6×6 (number of nodes $\times$ 2 DOFs at a node).

Boundary conditions and linear solution

After eliminating rows and columns of the global stiffness matrix and external force vector, global displacements can be solved:

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$

A size of $\mathbf{K}$ is 3×3 (all nodal displacements (DOF's) - three constraints = 3).

Member forces

Firstly you need to define global displacements for all elements. For example for element 1:

$$\mathbf{u}_G^{(1)} = \begin{pmatrix} u_{X_1} \\ \theta_{Z_1} \\ u_{X_2} \\ \theta_{Z_2} \end{pmatrix}$$

Then you need to present those in the elemental level. In this case, the element's local coordinate systems are parallel with the global coordinate system so no need for transformation i.e. $\mathbf{u}_G^{(i)} = \bar{\mathbf{u}}^{(i)}$. Member forces (for an element i) can be found as

$$\bar{\mathbf{f}}^{(i)} = \bar{\mathbf{K}}^{(i)} \bar{\mathbf{u}}^{(i)} \quad (14)$$